Test-Free Student Assessment

Bringing data and knowledge to the education of our students

ABSTRACT

A Deeper View has developed a software platform that fundamentally changes the way performance, problems and accomplishments of K-12 students, including special needs students, are measured and assessed.

With ADV Tracking, a teacher can capture detailed progress markers and indicators … and build them into long-term analytical records of individual student progress.

More importantly, a teacher can make the observations in the short time intervals where they normally observe students in classroom activity … opportunistically grabbing data with the click of a button.

ADV Tracking retains, organizes and presents this data for review and analysis, now and in the future.

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Problem Statement

In the current state of education in America, student performance is largely measured with after-the-fact, high stakes exams that are administered, at most, several times per year. The heavy consequences of these infrequent tests are believed to have led to highly dysfunctional activity, involving students, teachers and administrators. Notable examples include the No Child Left Behind initiative that has been widely condemned and, in the extreme case, the Atlanta Public Schools cheating scandal.

There are several forces that have fueled the drive to do more standard testing:

- Standardized tests are "standard". Whether they are accurate or not, they apply equally to every student who takes the test.
- Standardized tests are relatively inexpensive … especially if they are multiple choice.
- Standardized tests produce a few simple numbers as outputs … easy to record, easy to compare, and easy to use as management criteria.

But students are individuals … and anything but standardized!

Even people who value the above-stated benefits of standardized testing do not believe that the tests can adapt to differences in individual characteristics.

Why don't we measure more often and measure on a more individualized basis? To date every assessment innovation that has been tried has been defeated by the toxic alignment of four factors:

- **Teacher Workload:** Everything else being equal, to measure more accurately, we should measure more often, use more open-ended questioning techniques and adjust measures on an individualized basis. Implementing such a system would produce a massive increase in workload for the teacher and the administrator.
- **Student Needs:** Expectations for teachers have increased and diversified. There are more special needs children, more English-as-a-second-language (ESL) children, more mandated topics and subjects, and more social and political flashpoints and minefields to navigate or avoid.
- **Budgets:** Most school jurisdictions are under heavy budget pressures. Whatever their challenges, they are lucky if their funding stays stable, let alone increases.
- **Larger Class Sizes:** In many schools, especially those with difficult demographics and financial challenges, class sizes have increased.

These factors combine and compound to make any move to more frequent, individualized assessment prohibitively difficult.

So we are currently stuck with standardized testing.

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The Response – ADV Tracking from A Deeper View

A Deeper View (ADV) has deployed a new way to measure student performance. The ADV Tracking system is partially derived from insights in the KIHd System that was developed at George Mason University's Kellar Institute for Human disabilities, under the direction of Dr. Michael Berhmann. ADV has extended those concepts into a new system design to address four factors that have defeated previous student measurement initiatives.[1]

Teacher Workload

The ADV Tracking system takes the task of measuring student performance and spreads it out over the entire school year. In addition to the big standard tests, teachers can record meaningful observations spontaneously in the course of their normal teaching days … with minimal extra effort and minor disturbance to their teaching workflows.

It will take this child 20 sec to solve the problem

It will take this teacher 15 sec to record 3 observations about his performance

Figure 1 - There is time

By spreading the task thinner and wider, teachers can do more of it without added stress. In fact, by measuring more finely and more often, teachers can reduce their current workload by reducing note-keeping, spotting issues earlier, intervening more effectively and tracking remedial outcomes more closely.

Differing Student Needs

The ADV Tracking system is highly customizable … to the level of setting specific performance targets for individual students. Once individualized performance targets are entered into ADV Tracking (a fairly easy, one-time effort at the start of term), it can be refined and adjusted throughout the year. Flexibility is built into the

[1] A Deeper View previously licensed the KIHd System from George Mason Intellectual Properties, but has extended the design to the point that, by mutual agreement between ADV and George Mason, the ADV Tracking System is now an independent system design, solely owned by A Deeper View.

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observation recording process. There is no need to pre-plan or choreograph it as a separate classroom activity.

With individualized performance criteria, ADV Tracking can be configured to measure performance against a wide variety of criteria: against the Common Core Standard, against the personalized study plan of a special needs child, or against stretch goals established for a gifted student.

Budgets

ADV Tracking is a cloud-based web technology. It is delivered through familiar end-user devices such as computers, laptops, tablets and smartphones. A teacher can use several devices simultaneously or switch from one to another as convenience dictates.

Consequently, ADV Tracking has a stable overhead and modest usage costs. ADV can offer pricing that is attractive to even the most cash-strapped schools, teachers, home schooling parents, and parents with special needs children.

Class Size

ADV Tracking does not alter class size, but it is relatively insensitive to it. A teacher notes observations about students whenever an opportunity occurs. A larger class may mean a bit less teacher attention and less measurement detail for a given individual, but, over time, ADV Tracking will still collect enough fine-grained continuous measures on individual child performance. If anything, the teacher will end up with a better record of students' performance for far less effort. Teachers can review their own observations about every student. In this small way, ADV Tracking may actually make it a bit easier for teachers to cope with larger classes.

By overcoming all four resistance factors, ADV Tracking is a truly disruptive technology. It is a research-based tool with the potential to break the standardized test stranglehold on American education.

One last thing. Because of its ease of use and flexibility, ADV Tracking can equally be used to create and apply standard measures … in combination with individualized targets and observations. There is no reason to give up the comparability or consistency of standards. With ADV Tracking, you can have your cake and eat it.

So What is ADV Tracking?

ADV Tracking is a Real-Time Classroom Data Collection System

ADV Tracking is a structured data collection tool. It has a web server application that works with remote apps and clients, running on PCs, laptops, tablets and smartphones. A teacher can use a remote app to input observations about individual student performance during normal classroom activity.

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As Figure 1 showed, teachers often have small segments of time to observe individual children. Twenty seconds here, 15 seconds there. Maybe another few minutes when they are observing project or group activity (Figure 2).

These students will be working on the task for 10 or 15 minutes. The teacher will be chatting with them throughout.

It will take this teacher 20 sec to record a breakthrough in a student's understanding.

Figure 2 - More time in class activities

Today, in these short intervals, teachers are observing, assessing and analyzing student's performance, reactions, enthusiasm … and a host of other indications that good teachers read as easily as breathing.

But what happens to all of that expert insight and analysis? Probably something like the process in Figure 3.

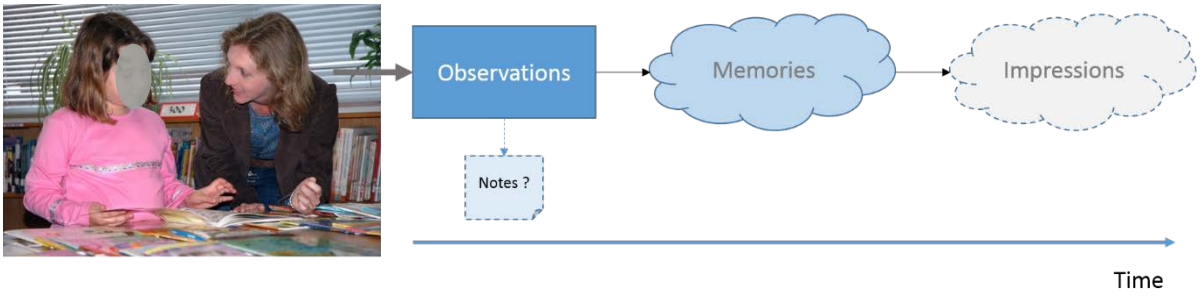

Figure 3 - Memories Fade

Except for a few, cryptic notes, most of the details of observation and analysis will quickly fade to memory, and eventually, as memories fade, be reduced to mere impressions. Those are valuable, but so much has been lost!

Expand this to tracking the progress of a student over a multi-month course (Figure 4). The vast majority of the teacher's expert observations are lost forever. All that remains are a few recollections and impressions.

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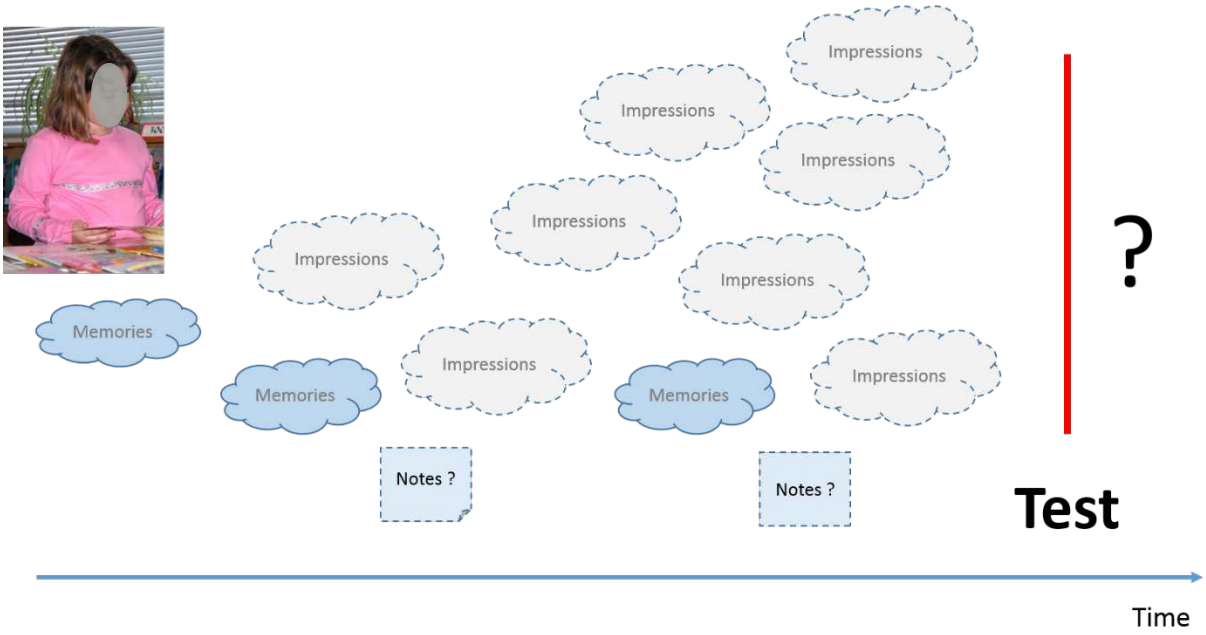

Figure 4 - So Little Remains

What do you usually hear at a parent-teacher meeting? You hear about the test results. When major tests are administered, the fruits of extended expert observation and interaction are abruptly replaced by the test scores. Even if the teacher knows other things about the student, it doesn't matter. Once the test results emerge, they take precedence. In today's world, subjective recollections and impressions can never compete with "hard" measurements.

Many of the best teachers try not to let this happen. They keep detailed records, journals and worksheets to retain more detail. But that activity takes time and effort. Time and effort that the teacher must steal from other activities … from their off-hours … from sleep.

ADV Tracking understands Privacy and Security

The topic of ADV Tracking's security architecture is beyond the scope of this document. However, ADV Tracking pays close attention to security of student data in its design and implementation. It is hosted on Microsoft web server technology that runs in Amazon Web Services. Both are widely respected platforms for sensitive applications.

ADV Tracking is a Disruptive Change

With ADV Tracking, a teacher can capture progress markers and indicators … and find them again. They can accumulate never-to-be-forgotten "observations" so they do not fade to impressions. Recorded observations can build into a sophisticated, long-term timeline of individual student progress.

The teacher can capture this information without sacrificing their family time or sleep. They can tuck the observations into intervals that occur naturally in classroom activity.

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ADV Tracking collects, retains, organizes and presents this data for instant review at any time … in quiet times … before parent meetings … even during class in the moment.

With ADV Tracking, teachers can now <u>afford</u> to do data-driven teaching!

How does ADV Tracking know what to record?

To make data collection practical in the moment, ADV Tracking uses pre-built measurement criteria that the teacher can select or activate with the press of a button. They have time for that, but not enough time to write text notes – especially on a mobile device.

So how does ADV Tracking present the right measures and choices?

At the heart of ADV Tracking there is a sophisticated, user-defined, multi-level rubric and data collection structure. This structure is built around 4 basic units:

- **Target** – A desired skill or knowledge mastery objective
- **Secondary Behavior** – other, non-task behaviors that may have occurred in combination with performance
- **Reinforcers** – What the teacher did to spur improvement.
- **Performance** – Level of skill or achievement
- … plus other options, attributes and choices to refine the inputs and results.

A person … a teacher, a consultant, a department head, or even a parent (home-schooling?) … defines the measurement rubric in the ADV Tracking system.

The rubric structure only needs to be input once. After that, it can be copied across students, across classes, across schools, across … anything. Once copied, it can be further tweaked to tailor it for individual students. In practice, this means that most of the observations collected by ADV Tracking are done using criteria that ***<u>is</u>*** standard across most or all students.

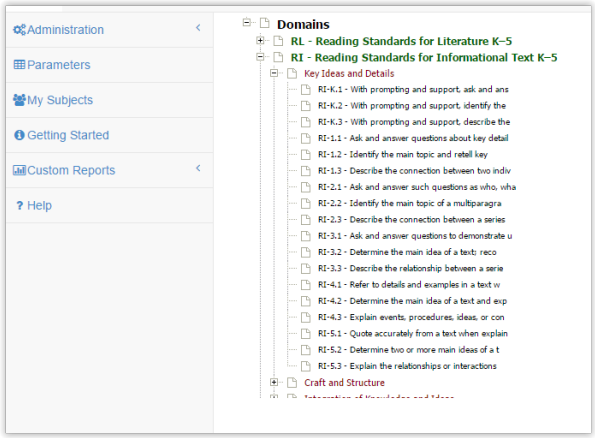

Figure 5 - Editable, Customizable Rubrics

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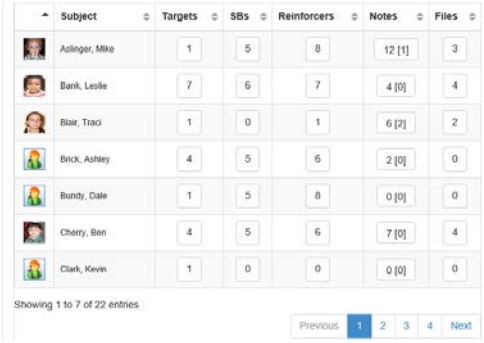
Finally, the task of selecting a suitable rubric is rapidly becoming easier. There is a national effort underway to build a standardized framework to topics and performance in K-12. The most comprehensive effort is the now-published "Common Core Standard". ADV includes a "starter rubric" based on the Common Core for every system. Anyone can start using ADV Tracking quickly by using it as a starting point and adding edits.

As more "standard" curricula are published, ADV will add the associated rubrics to the ADV Tracking library.

How does ADV Tracking work in the Classroom?

To the teacher, ADV Tracking is just a few screens and buttons on a smartphone or tablet.

- Pick a student and a target (Figure 5)
- Pick a sub-target or behavior
- Record performance, plus reinforcing & secondary behaviors. (Figure 6)
- Done!



Subject	Targets	SBs	Reinforcers	Notes	Files
Adlinger, Mike	1	5	8	12 [1]	3
Bank, Leslie	7	6	7	4 [0]	4
Blair, Traci	1	0	1	6 [2]	2
Brick, Ashley	4	5	6	2 [0]	0
Bundy, Dale	1	5	8	0 [0]	0
Cherry, Ben	4	5	6	7 [0]	4
Clark, Kevin	1	0	0	0 [0]	0

Showing 1 to 7 of 22 entries

Previous 1 2 3 4 Next

Figure 6 - Selection Screen

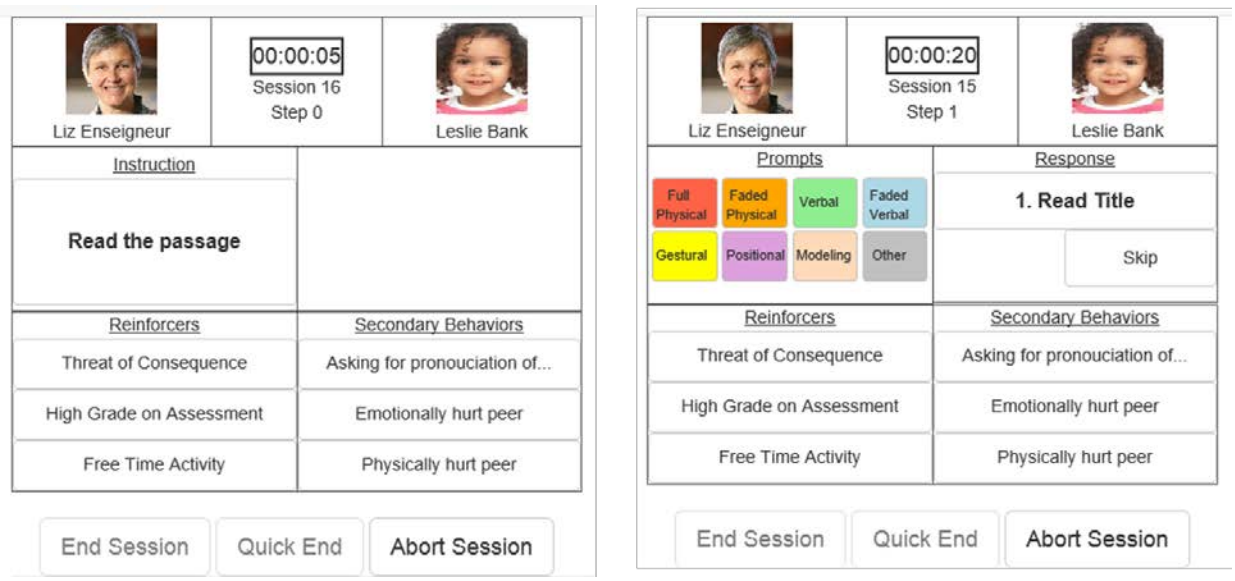


Figure 7 - Active Recording[2]

[2] Characters are fictitious.

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Teachers take notes now … but too often with a pen and paper!

These students will be working on the task for 10 or 15 minutes. The teacher is showing examples and taking notes **with a pencil and paper**. Where does that data go?

These students will be working on the task for 10 or 15 minutes. The teacher is showing examples and taking notes **with KIHd**. It goes into an analytical database.

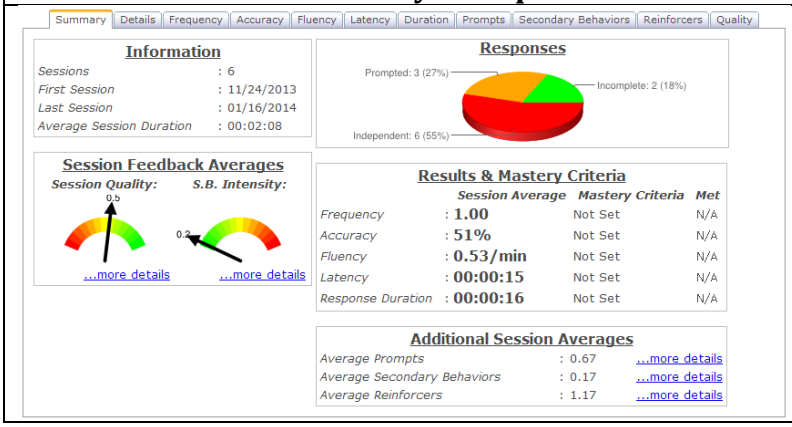
How much more data can they collect? How many meaningful measures can they extract? How much complex behavior can they record?

ADV Tracking is a Data Analysis System

It is a waste to collect a lot of detailed data … unless you can analyze and visualize it.

ADV Tracking has a suite of standard reports and the ability to construct custom reports and export data for analysis in other tools. Table 1 shows a selection from the standard reports and graphs.

Table 1 - Selected Standard ADV Tracking Reports

Standard Analysis Reports	Purpose
	All of the key data for performance against a target.

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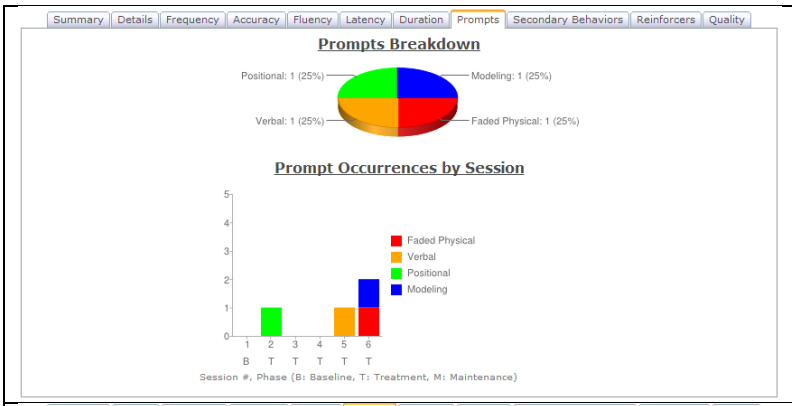
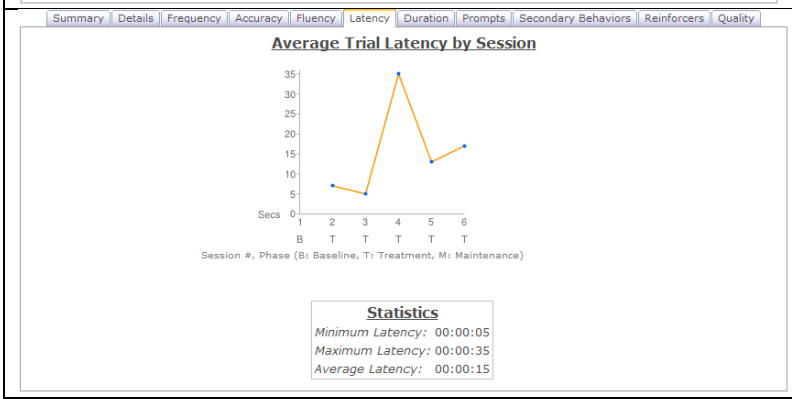
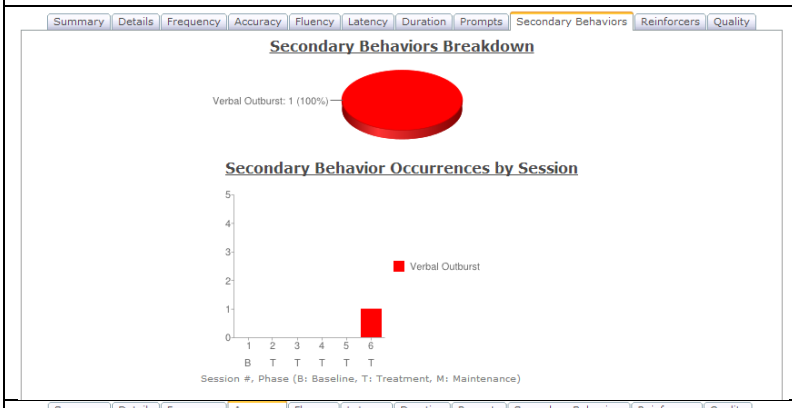
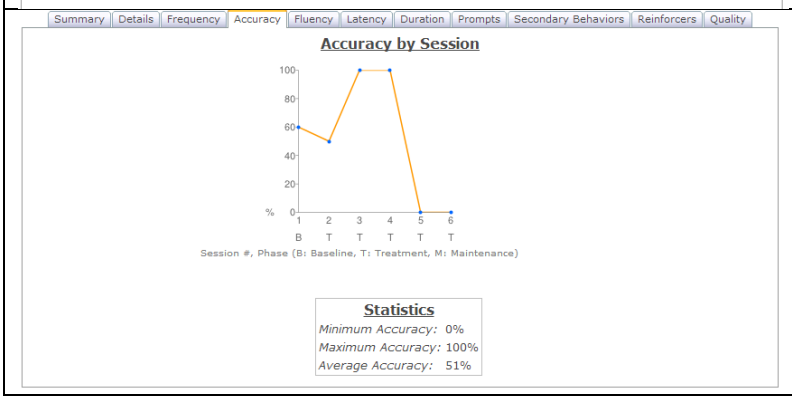
	A summary of the things that the teacher did to influence performance.
	ADV Tracking collects all of the observations with detailed timestamps. That means you can study both performance and the **"timing of performance".**
	A summary of the other things that the student did that might influence performance.
	A summary of accuracy of the performance against the target … over time.

The graphic representations meet the requirements for Response-to-Intervention (RtI) plans. The detailed data can be aggregated into summary data for classes, sections, class project teams, schools, school districts … possibly even states and nations.

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Imagine what sort of analyses will be possible with this level of detailed, time-stamped, target-specific performance data? What sorts of questions can it answer? Consider:

- Which types of prompts and reinforcers work best for skills involving memorization? e.g., multiplication?
- How long does it take between introducing a topic and 80% of students achieving its mastery?
- Which secondary negative behaviors are causing the biggest delays in learning?

These and questions yet to be imagined can be asked about individual students, grade years, schools, departments, states …

ADV Tracking can be Shared

ADV Tracking runs from the "cloud". If you have a valid account, you can use a client on your phone or tablet for any student that you are authorized to observe. ADV Tracking can assign different roles to different individuals to administer, design, collect data and review reports.

In a general teaching situation, authorized teachers, in different periods, can collect and share information in a consistent and transparent way.

For children with special needs, teachers could enlist the active support of parents and possibly consultants and therapists. Each can see the student's activities in a consistent, organized way.

In these cases, people will not be working cross purposes. Instead, all of the student's stakeholders can now reinforce everyone's plan and work.

ADV Tracking gives teachers fair accountability

Teachers can now be measured by the targets they set for each student … and the progress that each student achieved against those targets. Since both aspects are defined and measured, both can be reviewed in detail and critiqued or defended … with data.

Where can ADV Tracking be used?

ADV Tracking can be used wherever better, more continuous, performance measurements would help educational outcomes. Some examples for consideration:

- **Children with Autism Spectrum Disorder** – Helping these children and students was one of the primary motivations for the development of ADV. These children cannot communicate their feelings or needs easily. Often the only way to "understand" them is to observe their behaviors over an extended period … to get the insights to craft a program that can reach the students in a way they can understand. ADV Tracking is ideal for that role.

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- **Special Needs Children** – In virtually every school, teachers and administrators are required to develop … and apply … an individualized Education Action Plan (EAP) for each special needs student. Writing it isn't too hard. Applying it is a chore. If you teach each child differently, then you have to measure them differently. If you do it right with conventional tools, it's a paperwork nightmare. ADV Tracking makes it simple and easy.
- **Home Schooling** – Many parents feel that they have to teach their children at home, but they want them to stay on pace with their peers. How can they know? ADV Tracking opens up new opportunities to share information (with parent-controlled safeguards) … even for physically isolated students.
- **Schools with high at-risk populations** – Many schools are struggling to compete in a world of standardized tests … because they have one or more hands tied behind their back. Large percentages of at-risk students are taught in poor facilities and their education is supported by inadequate budgets. ADV Tracking can't reduce those challenges, but it can capture the data that **proves** how students' performance is actually changing. When you capture so many data points on so many students over an extended time, there is a wealth of data on what works and doesn't work … it is pretty hard to manipulate the outcomes … ***or criticize their validity!***
- **Teaching to the Common Core Standard** – As schools adopt the Common Core Standard, ADV Tracking becomes even easier to use. Most of the rubric is already there. Add a few edits and tweaks and every teacher is set up to capture detailed real-time data with minimal effort.

If you can capture valid detailed measures of student performance with minimal effort, why would you **not** want to do it?

Summary

ADV Tracking has the ability to "alter classroom instruction at the molecular level". It changes:

… and it doesn't increase teacher workload, straightjacket students into rigid categories, increase education budgets, or break down with larger classes.

In fact, ADV Tracking **can reduce teacher workload**. Teachers no longer have to maintain their own detailed records. They can spend their time using their own real-world data to develop more effective teaching methods.

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ADV Tracking can measure students against **individualized criteria, but store the results in a standard format**. It takes the same time to record an observation against an individual target as it does against a standardized one.

ADV Tracking costs as little $7.50 per student per year[3]. It can be recovered many times over by the operational savings that better data can generate.

ADV Tracking uses the brief intervals when teachers take a moment to observe. That may happen more in smaller classes, but **there are plenty of opportunities in larger classes**.

With ADV Tracking, teachers can now afford to do data-driven teaching!

For more information:

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[3] This price applies to site licenses for schools with large student populations. Individual and small unit pricing is higher, but will not exceed $50 per year.

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